Physics (PHYS-1101)

MECHANICS

Lecture 6:

Linear Momentum and Collisions

Momentum

- The linear momentum $\vec{\mathbf{p}}$ of an object of mass $\boldsymbol{m}$ moving with a velocity $\vec{\mathbf{v}}$ is defined as the product of the mass and the velocity.
  - $\vec{\mathbf{p}} = m\vec{\mathbf{v}}$
  - SI Units are *kg.m/s*
  - Vector quantity, the direction of the momentum is the same as the velocity's.

Momentum components

- $p_x = mv_x$ and $p_y = mv_y$
- Applies to <u>two-dimensional motion</u>.

Exercise

An object of mass 500g accelerates uniformly from rest at $4ms^{-2}$. Calculate its momentum at time t =6s.

Soln =<u>12 kg.m/s</u>

Impulse

- In order to *change* the momentum of an object, a force must be applied.
- The time rate of change of momentum of an object is equal to the net force acting on it.
  - $$\frac{\Delta \vec{\mathbf{p}}}{\Delta t} = \frac{m(v_f - v_i)}{\Delta t} = \vec{\mathbf{F}}_{net}$$
  - Gives an alternative statement of Newton's second law.

Impulse cont.

- When a single, constant force acts on the object, there is an **impulse** delivered to the object.
  - $\vec{\mathbf{I}} = \vec{\mathbf{F}} \Delta t$
  - $\vec{\mathbf{I}}$ is defined as the *impulse*
  - Vector quantity, the direction is the same as the direction of the force.

Impulse-Momentum Theorem

- <u>The theorem states that the impulse acting on the object is equal to the change in momentum of the object</u>.
  - $\vec{\mathbf{F}}\Delta t = \Delta\vec{\mathbf{p}} = m\vec{\mathbf{v}}_f - m\vec{\mathbf{v}}_i$
  - <u>If the force is not constant</u>, use the ***average force*** applied.

Example

A 150 N resultant force acts on 300 kg trailer. Calculate how long it takes this force to change the trailer's velocity from $2ms^{-1}$ to $6ms^{-1}$ in the same direction.

Solution:

$$m = 300kg, F_n = 150\ N, u = 2ms^{-1}\ and\ v = 6ms^{-1}$$

$$F_n.\Delta t = \Delta P$$

$$F_n.\Delta t = m(v - u)$$

$$\Delta t = \frac{m(v-u)}{F_n}$$

$$\Delta t = \frac{300kg(6ms^{-1} - 2ms^{-1})}{150\ N} = 8s$$

Impulse Applied to Auto Collisions

- The most important factor is the collision time or the time it takes the person to come to a rest.
  - This will reduce the chance of dying in a car crash.
- Ways to increase the time
  - Seat belts
  - Air bags

Air Bags

- The <u>air bag increases the time of the collision.</u>
- It will also <u>absorb some of the energy from the body.</u>
- It will **spread out the area of contact.**
  - decreases the pressure
  - helps prevent penetration wounds

Conservation of Momentum

- Momentum in an isolated system in which a collision occurs is conserved.
  - A collision may be the result of physical contact between two objects.
  - “Contact” may also arise from the electrostatic interactions of the electrons in the surface atoms of the bodies.
  - An isolated system will have not external forces.

Conservation of Momentum, cont

- The principle of conservation of momentum states <u>when no external forces act on a system consisting of two objects that collide with each other, the total momentum of the system remains constant in time.</u>
  - Specifically, the total momentum before the collision will equal the total momentum after the collision.

Conservation of Momentum, cont.

- <u>Mathematically</u>:

$$m_1 \vec{\mathbf{v}}_{1i} + m_2 \vec{\mathbf{v}}_{2i} = m_1 \vec{\mathbf{v}}_{1f} + m_2 \vec{\mathbf{v}}_{2f}$$

  - Momentum is conserved for the *system* of objects.
  - The system includes all the objects interacting with each other.
  - Assumes **only** internal forces are acting during the collision.
  - Can be generalized to any number of objects.

Notes about a system

- Remember conservation of momentum applies to the *system*.
- Momentum is a *vector quantity* therefore **direction** is important.
- Make sure you have the *correct signs* of the quantities.

Types of collisions

- There are generally <u>three</u> types of collisions namely;
  1. elastic collisions
  2. inelastic collisions
  3. glancing collisions

Types of collisions: Elastic Collisions

- Both momentum and kinetic energy are conserved.
- Typically have *two unknowns.*

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

$$\frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2$$

- Solve the equations simultaneously.

Elastic Collisions, cont.

- A simpler equation can be used in place of the KE equation.

$$v_{1i} - v_{2i} = -(v_{1f} - v_{2f})$$

Example

Two billiard balls of identical mass are moving towards each other. If the initial velocities of the balls are +30cm/s and -20cm/s, what are the velocities of the balls after the collision assuming they underwent a perfectly elastic collision

Solution

Write an equation for the conservation of momentum

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

Solution cont.

$$v_{1i} + v_{2i} = v_{1f} + v_{2f}$$

$$\frac{30cm}{s} + \left(-\frac{20cm}{s}\right) = v_{1f} + v_{2f}$$

10cm/s = $v_{1f} + v_{2f}$... (1)

Lets now apply the simpler version of the conservation of energy equation

$$v_{1i} - v_{2i} = -(v_{1f} - v_{2f})$$

30cm/s-(-20cm/s)= $v_{2f} - v_{1f}$

50cm/s = $v_{2f} - v_{1f}$ ... (2)

now solve the two equations simultaneously

Solution cont.

We get

v_{1f} = -20cm/s, $v_{2f} = +30cm/s$

NB: The balls have exchanged velocities, almost as if they had passed each other. This is always the case when to objects of the same mass have undergone a head-on elastic collision

Types of collisions: inelastic collisions

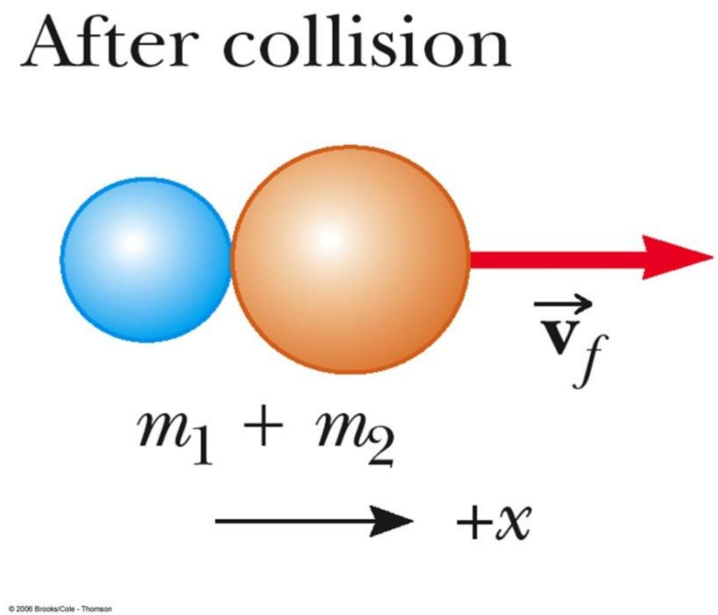
- Inelastic collisions
  - Momentum is conserved but kinetic energy is not conserved
    - Some of the *kinetic energy is converted into* other types of energy such as *heat, sound, work* to permanently deform an object.

Types of collisions: perfectly inelastic collision

- In a *perfectly* inelastic collision, momentum is conserved, kinetic energy is not.
- After the collisions, the objects stick together and move with a common velocity.
  - Not all of the KE is necessarily lost.

Types of collisions: Perfectly Inelastic collisions

- Conservation of momentum becomes



$$m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2) v_f$$

Problem Solving for One - Dimensional Collisions

- **Coordinates:** Set up a coordinate axis and define the velocities with respect to this axis.
  - It is convenient to make your axis coincide with one of the initial velocities.
- **Diagram:** In your sketch, draw all the velocity vectors and label the velocities and the masses.

Problem Solving for One - Dimensional Collisions, 2

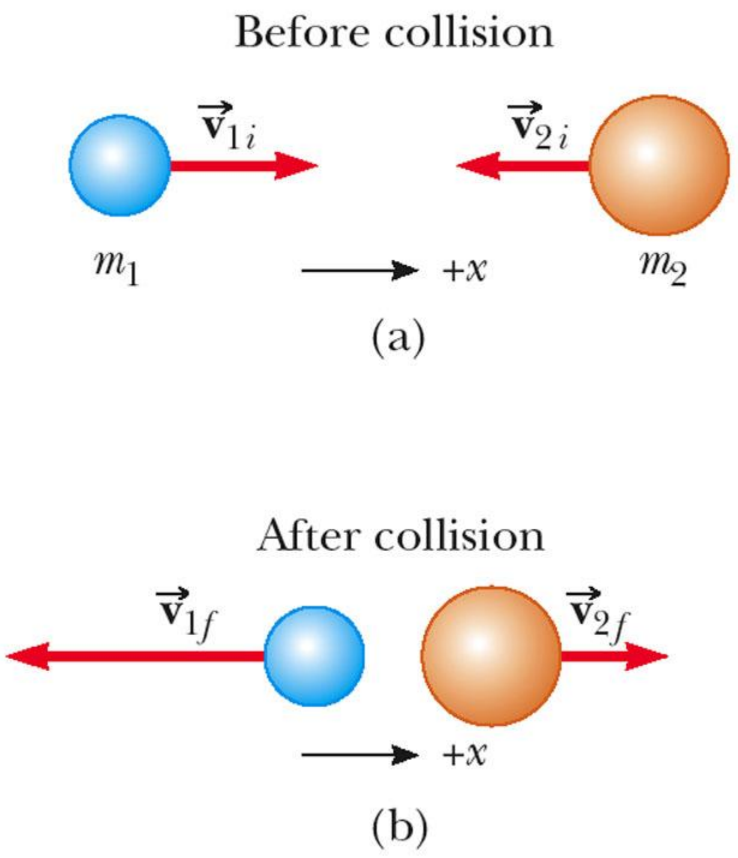
- **Conservation of Momentum:** Write a general expression for the total momentum of the system *before* and *after* the collision.
  - Equate the two total momentum expressions.
  - Fill in the known values.

Problem Solving for One - Dimensional Collisions, 3

- **Conservation of Energy:** If the collision is elastic, write a second equation for conservation of KE, or the alternative equation.
  - *This only applies to perfectly elastic collisions.*
- **Solve:** the resulting equations simultaneously.

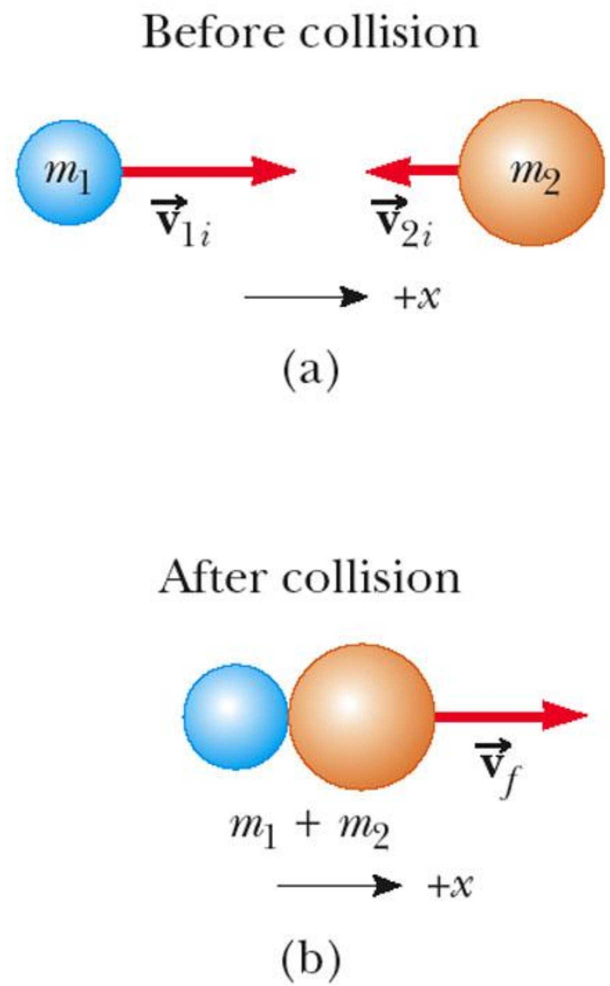
Sketches for Perfectly Elastic Collision Problems

- Draw "before" and "after" sketches.
- Label each object.
  - include the direction of velocity.
  - keep track of subscripts.



Sketches for Perfectly Inelastic Collisions Problem

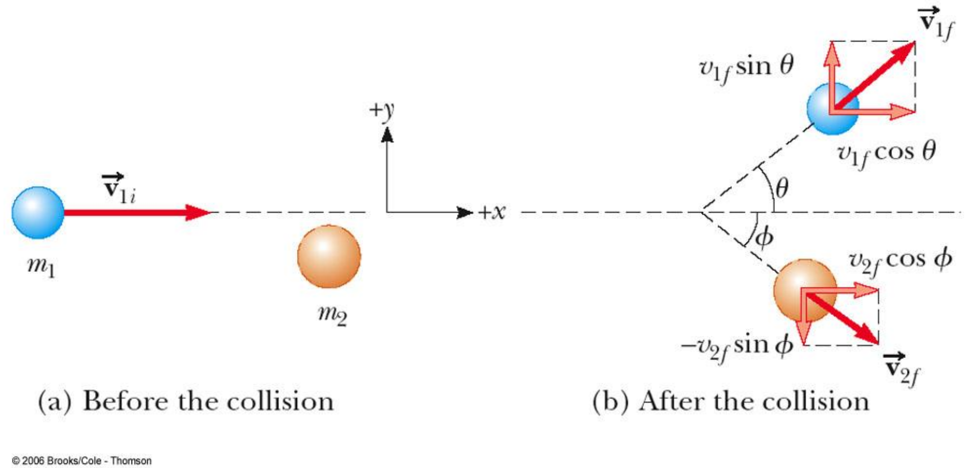
- After collision, <u>objects stick together</u>.
- Include <u>all the velocity directions.</u>
- The "after" collision <u>combines the masses.</u>



Types of Collisions: Glancing Collisions

- For a <u>general collision of two objects in three-dimensional space</u>, the conservation of momentum principle implies that the *total momentum of the system in each direction is conserved*.
  - $m_1 v_{1ix} + m_2 v_{2ix} = m_1 v_{1fx} + m_2 v_{2fx}$ and
    $m_1 v_{1iy} + m_2 v_{2iy} = m_1 v_{1fy} + m_2 v_{2fy}$
  - <u>Use subscripts for identifying the object, initial and final velocities, and components.</u>

Types of Collisions: Glancing Collisions

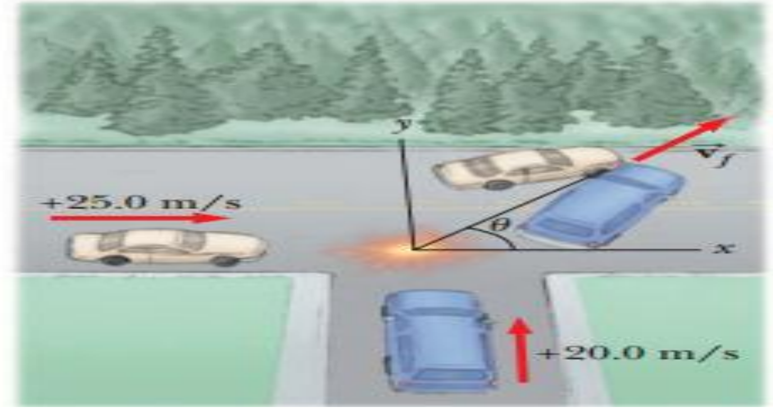


- The "after" velocities have x and y components.
- Momentum is conserved in the x direction and in the y direction.
- Apply conservation of momentum separately to each direction.

Example

A car with mass $1.50 \times 10^3 \text{kg}$ travelling east at a speed of 25.0m/s collides at an intersection with a $2.50 \times 10^3 \text{kg}$ van travelling north at a speed 20.0m/s. Find the magnitude and direction of the velocity of the wreckage after the collision assuming that the vehicles undergo a perfectly inelastic collision. Assume that the friction between the vehicles and the road is negligible

Solution



x- components:

$$\sum p_{xi} = m_{car}\, v_{car} = (1.5 \times 10^3 kg) \times 25 m/s$$

$$= 3.75 \times 10^4 \mathrm{kg.\,m/s}$$

$$\sum p_{xf} = (m_{car} + m_{van}) v_f cos\theta$$

Solution cont.

Equating initial x-component to final component,

$$3.75 \times 10^4 \mathrm{kg.m}/s = (4.0 \times 10^3 kg) v_f cos\theta$$

y- components

$$\sum p_{yi} = m_{van} v_{van} = (2.5 \times 10^3 kg) \times 20 m/s$$

$$= 5 \times 10^4 \mathrm{kg.m/s}$$

Solution cont.

$$\sum p_{yf} = (m_{car} + m_{van})v_f sin\theta$$
$$= (4.0 \times 10^3 kg)v_f sin\theta$$

Equating y-components

$$5 \times 10^4 \mathrm{kg.m/s} = (4.0 \times 10^3 kg)v_f sin\theta$$

Dividing y-component equation by x-component, we solve for θ

Solution cont.

$$tan\theta = \frac{5.0\times10^4 kg.m/s}{3.75\times10^4 kg.m/s}$$

$=1.33$

Therefore $\theta = 53.1^{\circ}$

And $v_f = \dfrac{5.0\times10^4 kg.m/s}{(4.0\times10^3 kg.m/s)\ sin53.1}$

$= 15.6 m/s$

Problem Solving for Two-Dimensional Collisions

- **Coordinates:** Set up coordinate axes and define your velocities with respect to these axes
  - It is convenient to choose the x- or y-axis to coincide with one of the initial velocities
- **Draw:** In your sketch, draw and label all the velocities and masses

Problem Solving for Two-Dimensional Collisions, 2

- **Conservation of Momentum:** Write expressions for the x and y components of the momentum of each object before and after the collision
- Write expressions for the total momentum before and after the collision in the x-direction and in the y-direction

Problem Solving for Two-Dimensional Collisions, 3

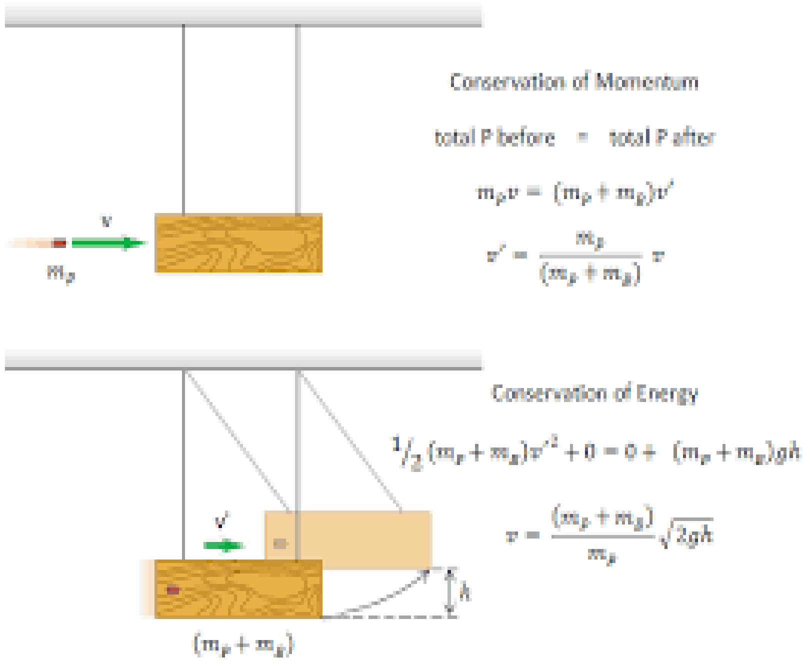
- **Conservation of Energy:** If the collision is elastic, write an expression for the total energy before and after the collision
  - Equate the two expressions
  - Fill in the known values
  - Solve the quadratic equations
    - Can't be simplified

Problem Solving for Two-Dimensional Collisions, 4

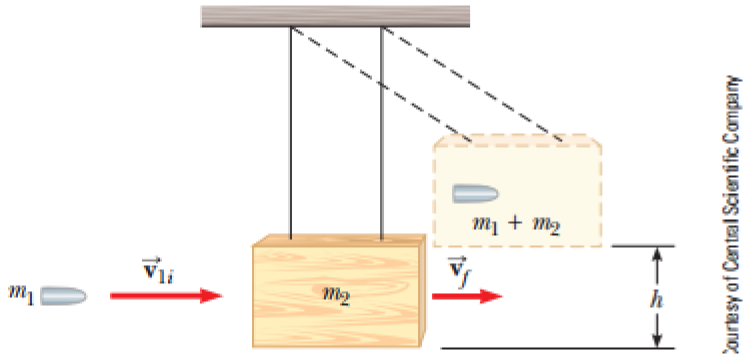
- **Solve** for the unknown quantities
  - Solve the equations simultaneously
  - There will be two equations for inelastic collisions
  - There will be three equations for elastic collisions

Ballistic pendulum

- Ballistic pendulum is a device for measuring the velocity of a fast moving projectile such as a bullet.
- The ***bullet*** is fired into a large block of ***wood*** which is suspended by some means of light wires.



Ballistic pendulum cont.



- The *bullet* is stopped by the *wood* and the entire <u>system</u> swings to a <u>**height *h***</u>

Ballistic pendulum cont.

- It is possible to obtain the initial velocity of the bullet if we know *h* and the two masses
- To do this, we use the conservation of momentum and perfectly inelastic collisions to find the initial speed of the bullet v_{1i} in terms of v_f (velocity after collision)

Ballistic pendulum cont.

- Secondly we use the the conservation of energy and height reached by the pendulum to find ϑ_f
- Then we substitute v_f into the previous result to find v_{1i}
- Mathematically put;

Ballistic pendulum cont.

- Conservation of momentum

$$p_i = p_f$$

$$m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2) v_f$$

- Conservation of energy

$$(KE + PE)_{after\ collision} = (KE + PE)_{top}$$

$$\frac{1}{2}(m_1 + m_2) v_f^2 + 0 = 0 + (m_1 + m_2) gh$$

Ballistic pendulum cont.

It then follows that, v_f^2 = 2gh

and $v_f = \sqrt{2gh}$

finally since $m_2 v_{2i}$ = 0, then

$$v_{1i} = \frac{m_1 + m_2}{m_1} \times \sqrt{2gh}$$

Example

In a ballistic pendulum experiment, the mass of the bullet, m_1 is 5.00g and the mass of the pendulum, m_2 is 1.00kg and h is 5.00cm. Find the initial speed of the bullet

Solution

Applying conservation of momentum

$$p_i = p_f$$

$$m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2) v_f \; \ldots \; (1)$$

$$(5.00 \times 10^{-3} \text{kg}) v_i + 0 = (1.005 kg) v_f$$

Applying conservation of energy

$$(KE + PE)_{after\ collision} = (KE + PE)_{top}$$

$$\frac{1}{2}(m_1 + m_2) v_f^2 + 0 = 0 + (m_1 + m_2) gh \ldots (2)$$

Solution

Therefore $v_f^2 = 2gh$

$$v_f = \sqrt{2gh} \text{ ... (3)}$$

$$= \sqrt{2\left(\frac{9.80m}{s^2}\right)(5.00 \times 10^{-2}m)}$$

$$=0.990 m/s$$

Finally, substituting this result into equation (1), we get

$$v_{1i} = \frac{(1.005kg)(\frac{0.99m}{s})}{5.00 \times 10^{-3}\ kg} = 199m/s$$

Exercise

1. A 70g bullet moving east at 400 m/s strikes a 1.2kg block on the right and becomes embedded in it. How high will the bullet-block system rise above its original point?

Thank You For Your Attention !!!